

Additional reading

For detailed information on sRGB, see the Microsoft and HP document "A Standard Default Color Space for the Internet - sRGB", available at www.w3.org/Graphics/Color/sRGB.html

Eco-friendly printing

According to a study by Lexmark, although consumers dispose cartridges responsibly, they take paper wastage casually

Feature

may not seem very much, but it is certainly good enough for most purposes – except professional graphics use, and that is where Adobe RGB came in. This is a larger colour space than sRGB, and was created by Adobe – the developers of graphics software such as Photoshop – to match the colour space that could be reproduced by most colour printers – a little less than double the number of colours of sRGB. Adobe sRGB is available for use in many higher-end digital cameras, and should be used if you plan on printing your images.

Using these two colour spaces, sRGB for use on general displays on the internet and Adobe RGB for colour printing, one can gain reasonably satisfactory results without the use of any calibration or specific profiles. But even these two can show up considerable differences, and one should check which is in use at any time. Consistency is vital. For example, load an image into Photoshop and then switch between colour spaces – the difference will be immediately obvious, and one image will be more true-to-life than the other.

However, hardware is improving with time, with camera sensors becoming more sensitive, monitors gaining much greater contrast range, and the gamut of printers increasing. These improvements, which show no signs of slowing down, give good reason for looking into more sophisticated forms of colour management than simply relying on sRGB and Adobe RGB.

As mentioned before, one major problem lies when you move from a device with one gamut, to a device with a smaller gamut. This is typically the case when printing. In a PC, an image file is held in RGB format. On your display, the varying values of red, green and blue combine to give the full range of colours. But printers work subtractively – you print onto a white surface, and subtract colour from it. The colour information is converted into a CMYK colour space – cyan, magenta and yellow – together with black. This subtractive process results in the printer gamut being smaller than other typical devices.

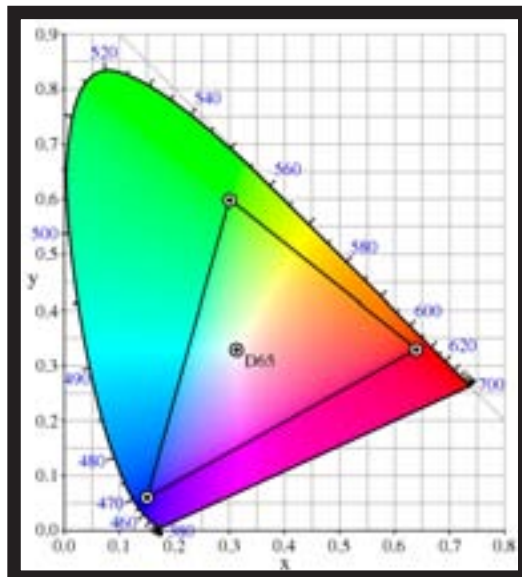
This is why Adobe took this colour space for the definition of Adobe RGB – its lowest common denominator. If you set your camera to use the Adobe colour space, a good quality PC and monitor can easily handle this, and little conversion is needed when printing – except from RGB to CMYK. This is a good choice for amateur photographers to print out their photos.

But the next step up entails some complications, and choices that need to be made. Perhaps the most important entails the concept of mapping "intent". There are three main methods, called colorimetric,

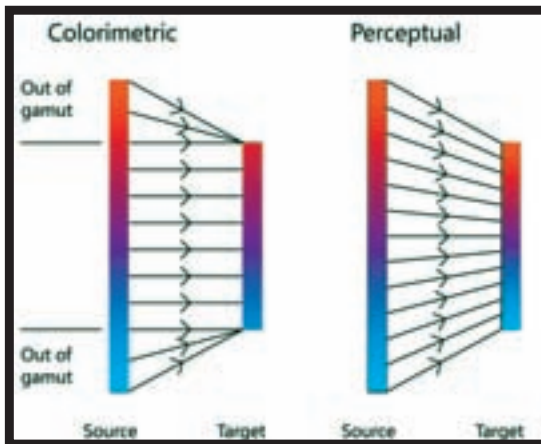
perceptual, and saturation. (We shall not cover saturation, which is most useful when going from a small colour space to a larger one.) Of the first of these, there are two variants, known as absolute and relative. The easiest to describe first is perceptual intent.

With perceptual intent, the colour range from the larger source gamut is scaled down to match the smaller target colour space. The "distance" between the colours is reduced, but their relative positions remain essentially the same. This method gives the best results from the point of view of human perception – hence the name. The two types of colorimetric intent handle the colours that are shared in both gamuts by mapping them directly – actually, there is a difference in the two variants here regarding this, with relative colorimetric not doing a direct mapping, but this need not concern us for now.

The important feature lies with those colours in the source gamut that lie outside of the target gamut – they are all simply mapped to the nearest colour point within the target gamut; what one



A 2D representation of the CIE XYZ standard colour space. It is intended to represent the entire gamut that human vision can perceive. The triangle overlaid on it is the approximate extent of the sRGB gamut.



A simplified 2D representation of mapping from one colour space to another which is smaller. On the left, with colorimetric intent, colours that are outside of the target gamut are mapped to the same colour at the edge of that gamut. On the right, with perceptual intent, the colour space is scaled down to fit the target gamut

might think of as the edge of that gamut. In this way several colours in the source colour space are all mapped to the same colour in the target. It is important to consider the colours that are actually in use in a particular image – many cameras and imaging software give different ways of displaying this information, and it is worth taking the time to understand this information.

If, for example, the great majority of the

colours in an image lie within the gamut of the target printer device, then using absolute colorimetric intent will be the most accurate – there is no point in scaling them down if they already fit, as you would do with perceptual intent. However, the greater the difference between the source and target gamuts, then the more important become those colours that lie outside of the target gamut. Think of an outdoor shot with much detail in green foliage, many different shades of dark green – colours that might well be outside of the printer gamut. Using colorimetric intent with this image would likely have most of those colours end up the same, losing all the detail. In that instance, perceptual intent would preserve more detail and yield a more satisfying result.

So, having described the main issues involved with colour management, and particularly with printers, we will now look at a practical example using the Spyder3Print.

Using the Spyder3Print

The main component of the Spyder3Print is the colorimeter that is used for taking measurements of colour patches that have been printed. There is also a guide that you lay down on the paper under inspection, a stand into which the device sits and also a CD containing software. The colorimeter is shaped rather like a large mouse, and the pointed end contains the sensor – just a few mms across – that is used for taking colour measurements. Unfortunately, there is zero documentation, and the user is expected to